

$$(1) \quad p(x) = 1 + x + x^4$$

$$d = 5$$

$$\alpha, \alpha^2, \alpha^3, \alpha^4$$

$$g(x) = (x - \alpha)(x - \alpha^2)(x - \alpha^3)(x - \alpha^4) =$$

$$= (x + \alpha)(x + \alpha^2)(x + \alpha^3)(x + \alpha^4) =$$

$$= x^4 + x^3(\alpha + \alpha^2 + \alpha^3 + \alpha^4) + x^2(\alpha^7 + \alpha^6 + \alpha^5 + \alpha^4 + \alpha^3) + x(\alpha^9 + \alpha^8 + \alpha^7 + \alpha^6) + \alpha^{10} \quad (=)$$

$$\alpha + \alpha^2 + \alpha^3 + \alpha^4 = 0010 \oplus 0100 \oplus 1000 \oplus 0011 = 1101 = \alpha^{13}$$

$$\alpha^7 + \alpha^6 + \alpha^5 + \alpha^4 + \alpha^3 = 1000 \oplus 0011 \oplus 1100 \oplus 1011 = 1100 = \alpha^6$$

$$\alpha^9 + \alpha^8 + \alpha^7 + \alpha^6 = 1100 \oplus 1011 \oplus 0101 \oplus 1010 = 1000 = \alpha^3$$

$$(=) \quad x^4 + \alpha^{13} x^3 + \alpha^6 x^2 + \alpha^{10}$$

② $PC(15, 9)$

$n=15$ $k=9$

$d=n-k+1=7$

$t=\frac{7-1}{2}=3$

корни

$g(x) = \alpha, \alpha^2, \alpha^3, \alpha^4, \alpha^5, \alpha^6$

$p(x) = 1+x+x^4$

$$\begin{aligned} v(x) = & \alpha + \alpha^3 x + \alpha^{10} x^2 + \alpha^{12} x^3 + \alpha^{13} x^4 + \alpha^{13} x^5 + \\ & + \alpha^9 x^6 + \alpha^5 x^7 + x^8 + \alpha^{10} x^9 + \alpha^2 x^{10} + \alpha x^{11} + \alpha^{12} x^{12} + \alpha^{13} x^{13} + \\ & + \alpha^2 x^{14} \end{aligned}$$

$S_j = v(\alpha^j)$

$j=1, 2, \dots, 6$

$$\begin{aligned} S_1 = & \alpha + \alpha^3 + \alpha^{10} + \alpha^{12} + \alpha^{13} + \alpha^9 + \alpha^5 + \alpha^{10} + \alpha^2 + \alpha + \alpha^{12} + \alpha^{13} + \alpha^9 + \\ & + \alpha^{12} + \alpha^{13} + \alpha^{11} + \alpha^1 = \alpha^2 + \alpha^{12} + \alpha^3 + \alpha^{13} + \alpha^{11} = \end{aligned}$$

$= 1111 \oplus 0100 \oplus 1000 \oplus 1101 \oplus 1110 = 0000 = \emptyset$

$$\begin{aligned} S_2 = & \alpha^2 + \alpha^5 + \alpha^{14} + \alpha^3 + \alpha^8 + \alpha^{13} + \alpha^6 + \alpha^4 + \alpha^1 + \alpha^{13} + \alpha^{13} + \\ & + \alpha^8 + \alpha^{10} + \alpha^9 + \alpha^0 = \alpha^5 + \alpha^{14} + \alpha^3 + \alpha^4 + \alpha^{13} + \alpha^8 + \\ & + \alpha^{10} + \alpha^9 + \alpha^0 = \end{aligned}$$

аналогично остальные синдромы 0

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \Lambda_3 \\ \Lambda_2 \\ \Lambda_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Вывод: так как синдром 0, то

ошибок нет, значит

вектор ошибок нулевой